FLEET TEAM DIGEST

737MAX-FTD-24-18001

Issue Title : P5-13 ELEC Light Illumination

Airplane Model	ATA	Minor Model(s)
737MAX	2434-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
10/05/2018	02/08/2024	01/24/2024

Estimated Completion Date	Next Update Date
04/28/2024	04/28/2024

Revision Description

Milestones updated to reflect the P91 Suppression Diode Design Change Honeywell Service Bulletin release and Production Incorporation.

Applicability

737MAX airplanes with:

Standby Power Control Unit P/N 1151952-19 (Honeywell) & P/N 510900412-001 (Honeywell)

Battery Charger P/N 8-930-03 (CRANE) & P/N 81-141-01 (CRANE)

Battery P/N BA35-01 (CRANE), P/N 024147-000 (SAFT),

Static Inverter P/N 1-002-0102-2170 (Avionic Instruments Inc.)

P5-13 panel P/N 285A1840-10, P/N 285A1840-9 (BAE Systems)

Description

Operators using the latest configuration of Standby Power Control Unit, Static Inverter, and Battery Charger continue to experience P5-13 ELEC light illuminations. Several reports indicate the annunciator can be cleared by BITE tests or manual resets. Test results of the removed units often determine to be no fault found (NFF).

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Background

Operators posted their fleet experiences on Boeing Fleet IdeaXchange (FIX), ISE-24-18-21969, to express concerns regarding nuisance P5-13 ELEC light illumination on 737MAX and 737NG. Some removed LRUs tested by suppliers were NFFs. Eighteen operators shared their experiences, in which one operator reported a maintenance success story in minimizing the nuisance alert.

Components related to the ELEC light illumination are: Standby Power Control Unit (SPCU), Battery Charger (BAT CHGR), Battery (BAT), Static Inverter (STA INV), and P5-13 panel.

Status

ELEC Light Overview

Boeing authorized Service Related Problem (SRP) 737MAX-SRP-24-0216 to investigate the Standby Power System and determine improvement opportunities and organized the Standby Power System Faults Working Together Team (WTT) in partnering with operators and suppliers to share experiences, maintenance practices and investigate the issue.

Boeing Analytics were reconciled with customer statistics for Elec Light task prioritization, held 7 WTT conferences with operators and suppliers (Avionic Instruments, CRANE, Honeywell), instrumented four airplanes (1 NG and 3 MAX) for on-wing ground testing and performed lab testing on the standby power system and components to identify any anomalies that could contribute to the issue. The WTT 6 and 7 presentation are attached in this FTD.

To investigate airplane power transfer anomalies that may cause ELEC light illumination, Boeing performed ground testing on 4 instrumented airplanes to measure various power system responses from SPCU, static inverter, battery chargers, and P5-13 panel. The ground test was used to verify a troubleshooting procedure for standby power system ELEC light issue. It was found that the power system response times were well within specifications.

The Elec Light Best Practices and recommendations documented in the WTT and FTM briefings has been released as a Best Practices Service Letter, 737-SL-24-238.

In parallel, to increase SPCU reliability and further reduce Elec Light Faults, the effort to release a SB for the P91 Suppression Diode Design Change has been initiated.

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Individual Elec Light component INOP status is contained in attachment to this FTD (Attachment 737MAX-FTD-24-18001 Status)

Interim Action

ELEC Light

For any ELEC Light illumination Boeing encourages operators to follow FIM 24-31 Task 801 to analyze the source of the ELEC Light signal and follow the associated BITE Procedure tasks.

Even if intermittent ELEC Light INOP is resettable at Gate Return, it is a Schedule Interruption. Boeing recommends operators maintain record/count of the intermittent ELEC Light INOP components for purposes of tracking of rogue units/rogue airplanes to facilitate if additional system scrutiny in necessary.

SPCU

Boeing encourages operators to follow FIM 24-34 Task 802: cycle C1410 - SPCU Normal and C1411 - SPCU Standby CBs to reduce Intermittent/NFF removals.

Static Inverter

Boeing encourages operators to follow FIM 24-34 TASK 801 to reset the static inverter power via CB C01343 before removal to minimize NFF removals for Static Inverter P/N 1-002-0102-2170, prior to SB 1-002-0102-2170-24-47 implementation.

Battery/Battery Charger

To improve a trend of Battery Charger or Battery NFF, Boeing encourages operators to follow FIM 24-31 Task 829: Utilize Charger front panel Lights to differentiate battery vs charger failure.

Final Action

Battery/Battery Charger

Boeing and CRANE were working together to identify any hard failures of the battery charger that could trigger ELEC Light alert/

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BAT CHGR INOP message. The investigation found resistor R14 of power input circuit failed in some returned units. The failures appear to be electrical stress on the resistor. The investigation determined the electrical stress could come from outside power sources and affected small percentage of chargers. CRANE has issued Service Letter 8-930-01-15 dated 01-APR-2020 to inform operators how to identify/correct the issue.

Crane SB 8 930 24 02: AC Filter Caps

Crane SB 8 930 24 03: IGBT CCA

Improved Configuration: P/N 81-141-01 (LN 7668, Mar 2021)

A Battery Discharge Check has been implemented into the 737MAX AMM, AMM Task 24-31-11-710-801, as of MAY-2021. This procedure contains steps to functionally check the main and if applicable, the auxiliary battery, to determine their state of charge.

(a) When batteries are found to be in a low state of charge, this procedure contains steps to evaluate them and will refer the operator to accomplish a battery installation test if necessary.

(b) When batteries discharge at an excessively high rate, this procedure contains steps to evaluate and will refer the user to the applicable fault isolation procedure.

(c) This procedure does not test the battery capacitance, battery capacitance is tested per CMM.

As of 15-NOV-2021, On-wing Ground Bond Checks at both the Battery and Battery Charger Grounds (higher than 8 AWG), have been completed. Boeing was able to confirm that all meters called out to test the ground bond for the battery and battery charger grounds are sufficient. Therefore all called out ground bond meters, Avtron T477W, Amptec 620LK and BCD Electronics M1/M1B are sufficient. No AMM, FIM or SWPM changes are necessary. The maximum allowed resistance for the battery and battery charger grounds is 0.1 milli-ohms.

Static Inverter

Boeing has also worked with suppliers in reviewing repair histories to identify trends and minimize ELEC light illumination.

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Investigations into the high rate of NFF returns of static inverter, P/N 1-002-0102-2170, found some power conditions may cause the inverter's output over-voltage alarm circuit to latch and trigger the ELEC light/STA INV INOP message, which required a power reset to clear. Boeing revised FIM 24-34 Task 801 in June 2019 to include cycling the Inverter Remote CB to clear the alert/ message. Avionic Instruments released SB 1-002-0102-2170-24-47 in July 2019 to modify the alarm circuit and prevent the erroneous alert/message.

For Static Inverter -2170, SB 1-002-0102-2170-24-42, Rev. 2 addressed S/N CK001566-CK001652 to relocate capacitors that may experience failure due to the location and also addressed S/N CK00122-CK001830 which replaces 3 capacitors that may not meet solder fillet height requirements.

For Static Inverter -2170, Service Information Letter (SIL) 1-002-0102-2170-24-48, Rev. 3 addressed moisture contamination of capactiors for S/N CK00120-CK003700. If incorporated the end of the serial number will be marked with -BD.

Boeing suggests operators to incorporate all SB's and SIL's to help eliminate Static Inverter related ELEC light illuminations.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Battery Health Check AMM (-8200): ECD 15-DEC-2023

Battery Health Check AMM (-7): ECD 15-DEC-2023

P91 Suppression Diode Design Change:

- Production Parts Available: ECD - January-2024 (Complete)

- Service Bulletin Release: ECD - 15-March-2024 (Target)

COMPLETED:

SPCU Rogue Unit Lab Testing: Completed 24-AUG through 28-AUG-2020, at Boeing Seattle Test Lab.

SPCU Rogue Unit On-wing Testing with Boeing Charlie Works designed Breakout Box (BoB): Completed 08-OCT and 09-OCT-2020, supported by SWA on a 737-700 aircraft.

Static Inverter Service Bulletin Release: Completed APR-2021

Battery Health Check AMM (-8/-9): Completed 15-MAY-2021

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WTT Meeting #6: Completed 14-JUN and 16-JUN-2021

SPCU Aux Contact Check CMM Addition: Completed 11-AUG-2021

FIM Release MAX: Completed 15-AUG-2021

Ground Bond Meter Testing Complete: Initial Testing Completed OCT-2021, Follow-up Testing Complete 15-NOV-2021

Working Together Team 7 (including ELEC Light Best Practices): Completed 25-OCT-2021

737 MAX OOS (including ELEC Light Best Practices): Completed 22-SEPT-2022

Boeing Authorization of P91 Suppression Diode Design Change: Completed 02-DEC-2022

ELEC Light Best Practices Service Letter (737-SL-24-238): Completed 13-JAN-2023

FIM Revision MAX: Adding excessive current battery discharge troubleshooting guidance and adding enhanced battery thermistor checks, Completed 15-JAN-2023.

AMM Revision MAX: Add NOTE on troubleshooting multiple failures and Battery Discharge Check Revision to add link to FIM for excessive battery discharge, Completed 15-JAN-2023.

SPCU Relay Survivability Assessment: Honeywell Service Information Letter 24-C-181 released May 2023

Operator Action

Boeing encourages operators to enter FIX postings of their experiences and maintenance practices that help minimize the issue. Also, Boeing encourages operators to attend WTT's, Fleet Team Meetings and Fleet Calls with ELEC Light agenda items to help in resolving the issue.

Part Information

Standby Power Control Unit P/N 1151952-19 (Honeywell) & P/N 510900412-001 (Honeywell)

Battery Charger P/N 8-930-03 (CRANE) & P/N 81-141-01 (CRANE)

Battery P/N BA35-01 (CRANE), P/N 024147-000 (SAFT),

Static Inverter P/N 1-002-0102-2170 (Avionic Instruments Inc.)

P5-13 panel P/N 285A1840-10, P/N 285A1840-9 (BAE Systems)

Reference Categories

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Reference Type	Reference Number	Revision Number	Reference Date
Service Letter (SL)	737-SL-24-238		
Other	FIX ISE-24-18-21969		05/21/2018
Service Related Problem (SRP)	737MAX-SRP-24-0216		

Related Categories

- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations

Attachments:

a) Working Together Team 7 - Meeting Slides

b) Attachment 737MAX-FTD-24-18001 Status

c) Working Together Team 6 - Meeting Slides

d) Lab and On-Wing Aux Relay Tests

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